Sahaj Bhadja

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Summary

Software engineer with 2+ years building distributed services, real-time integrations, and cross-platform systems. Experienced in C++, Python, TCP/IP, REST APIs, WebSockets, microservices, sensor-data workflows, algorithms, cloud infrastructure, automated testing, and production reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected Python API, Azure SQL, and Blob Storage services for 500+ users, refining indexed queries, secure interfaces, and access patterns to reduce distributed dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant asynchronous processing for 5,000+ monthly integration jobs using explicit states, retries, reconciliation, and failure isolation, tripling throughput while sustaining 99%+ reliable production behavior under concurrent load.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Flask REST services and integration workflows for secure ingestion, routing, status, and history tracking, translating external requirements into maintainable interfaces supporting 150+ users and 10,000+ daily requests.
- Created concurrent processing for 1,000+ files with automated validation, error handling, tests, and AWS RDS persistence, reducing turnaround 60% while maintaining stable application behavior during peak workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Integrated time-aligned LiDAR and vision sensor tokens into a spatial-temporal model, building reproducible simulation pipelines over 100,000+ records and improving long-horizon planning accuracy 18% across controlled evaluations.
- Participated in cross-functional collaborative design reviews and validation experiments, documenting technical decisions and tuning a spatial-temporal architecture that improved closed-loop stability 15% across repeated controlled simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed object-oriented API and ETL integration components using reusable design patterns, AWS Glue, and S3, improving subsystem response times 30% and cross-platform data-flow efficiency 68% across production workflows.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating defect diagnosis and improving engineering confidence in production releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed eight modular services with REST APIs and WebSocket messaging for authentication, resource allocation, permissions, and notifications, synchronizing 2 million+ Azure SQL records across independently maintainable subsystem boundaries.
- Automated Docker builds, tests, staged deployments, and production monitoring through Jenkins and Azure services, reducing release time 50% while improving overall software quality and consistent operational reliability.

Skills

Languages & Core Systems: C++, Python, Java, TypeScript, JavaScript, SQL, Bash, object-oriented programming, data structures, algorithms, concurrency, multithreading, performance optimization
Networking & Integration: TCP/IP, REST APIs, WebSockets, microservices, heterogeneous integration, API design, event-driven workflows, asynchronous processing, real-time synchronization, authentication, distributed systems
Data & Cloud: AWS, Azure, GCP, AWS RDS, S3, PostgreSQL, MySQL, MongoDB, Azure SQL, Blob Storage, data modeling, indexing, caching, query optimization
Quality & Delivery: Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, PyTest, Mockito, unit testing, integration testing, simulation testing, debugging, observability, monitoring, documentation, Agile, Scrum

Projects

CPU-Scheduling Simulator | Live | *C++, React, JavaScript, Algorithms*
- Implemented six scheduling algorithms with dynamic metrics and Gantt visualization across 1,000+ process combinations, reducing debugging time 20% through deterministic simulation, maintainable C++ logic, and systematic edge-case testing.

FinNexus | Live | *TypeScript, React, Tailwind CSS, WebSockets, Kubernetes*
- Built a real-time distributed platform supporting 50,000+ users and one million transactions, integrating secure APIs, WebSocket synchronization, advanced filtering, MongoDB persistence, Kubernetes, and automated deployment workflows.

Big-Brain | Live | *Python, PostgreSQL, GCP, REST APIs, FAISS*
- Engineered Python-oriented cloud ingestion and indexed workflows processing 50,000+ records with PostgreSQL, GCP, and REST APIs, implementing secure access, incremental updates, and performance tuning to reduce retrieval latency 40%.